

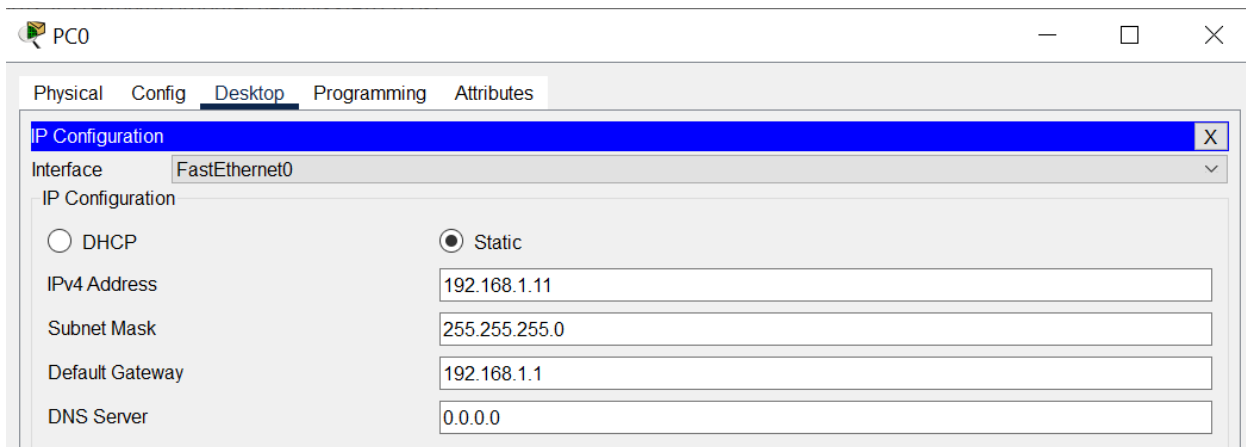
Configuring IP Addresses to PCs

- **PC0** → IP Address: **192.168.10.2**, Subnet Mask: **255.255.255.0**, Gateway: **192.168.10.1**
- **PC1** → IP Address: **155.168.1.2**, Subnet Mask: **255.255.255.0**, Gateway: **155.168.1.1**

OUTPUT:

PC IP configuration:-

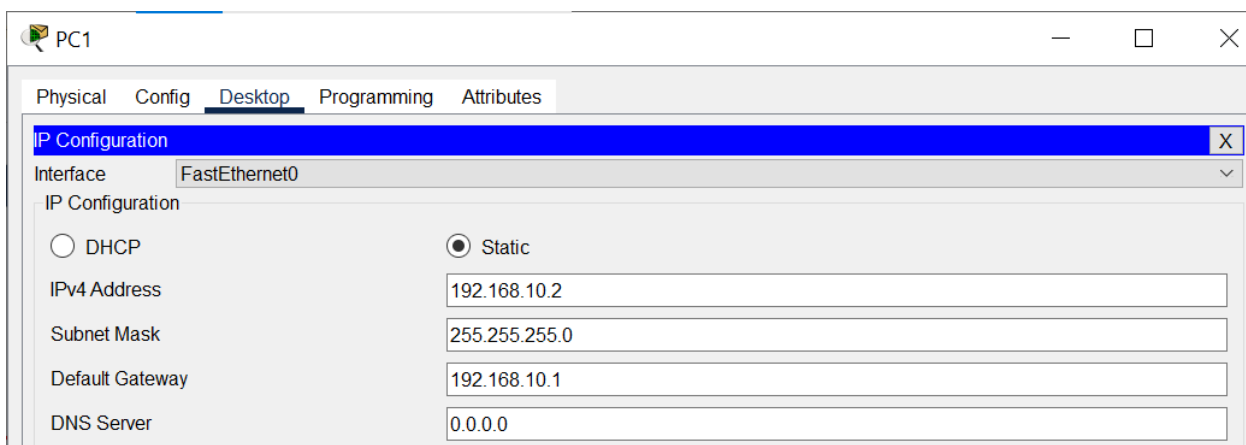
PC0/LAPTOP IPconfiguration



The screenshot shows the configuration window for PC0. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The fields are filled with the following values:

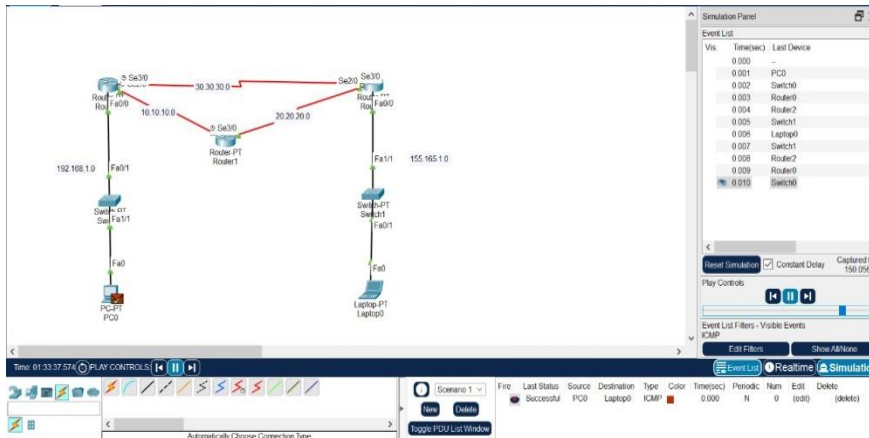
Field	Value
IPv4 Address	192.168.1.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

PC1 IPconfiguration



The screenshot shows the configuration window for PC1. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The fields are filled with the following values:

Field	Value
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0



Router Configuration:-

At ROUTER 0:-

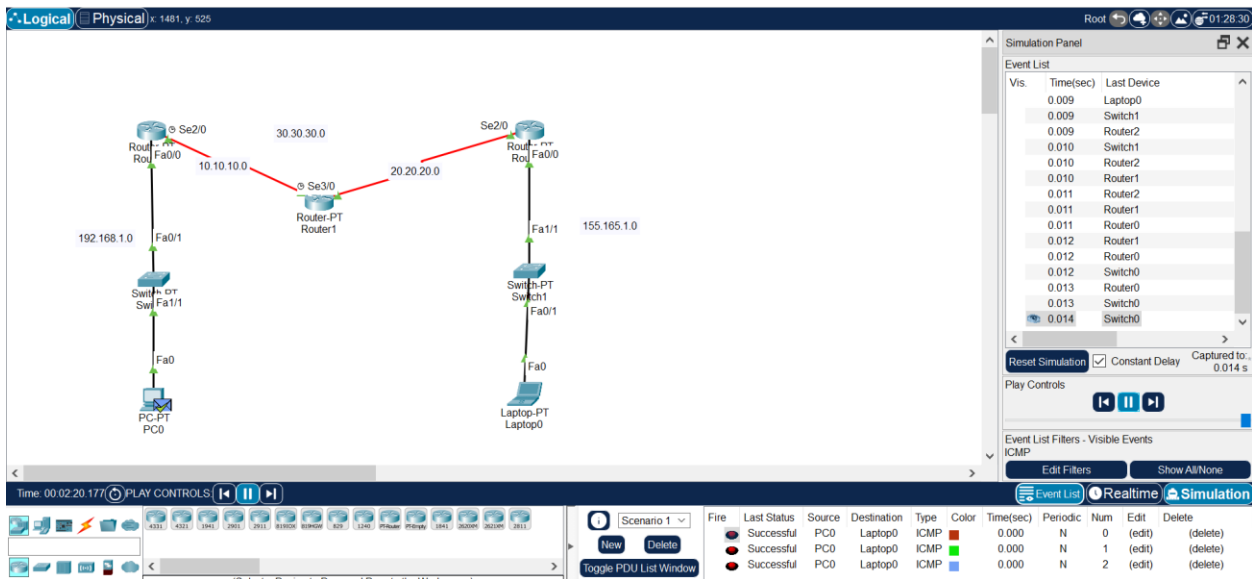
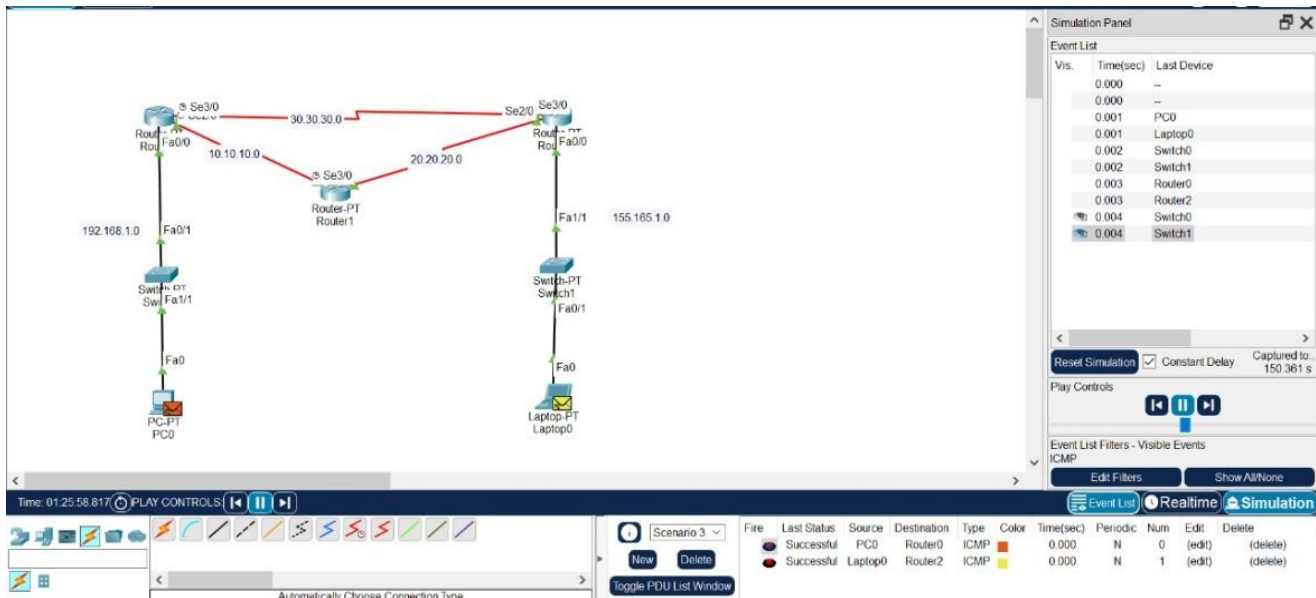
```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.255.255.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.255 area 0
Router(config-router)#network 30.30.30.0 0.0.0.255 area 0
Router(config-router)#exit
```

AT ROUTER 1:-

```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 10.10.10.0 0.0.0.255 area 0
Router(config-router)#network 20.20.20.0 0.0.0.255 area 0
Router(config-router)#exit
```

AT ROUTER 2:-

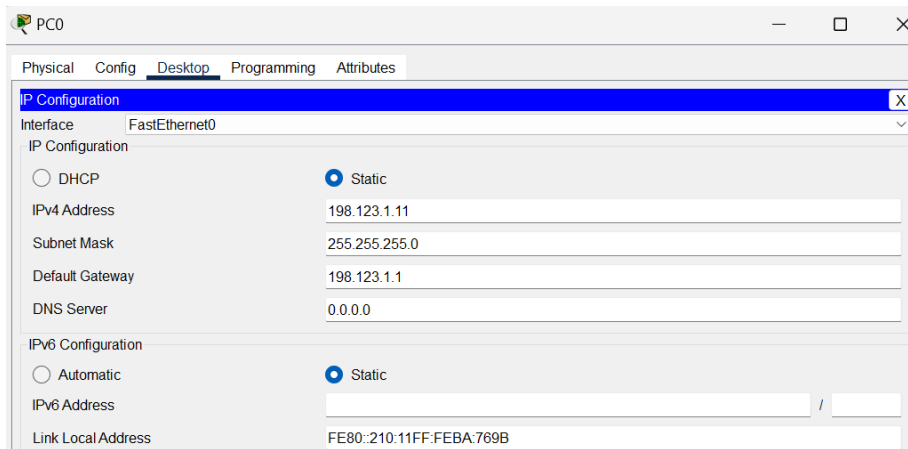
```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 30.30.30.0 0.0.0.255 area 0
Router(config-router)#network 155.165.1.0 0.0.255.255 area 0
Router(config-router)#exit
```

Final result:-

OUTPUT:

PC IP configuration:-

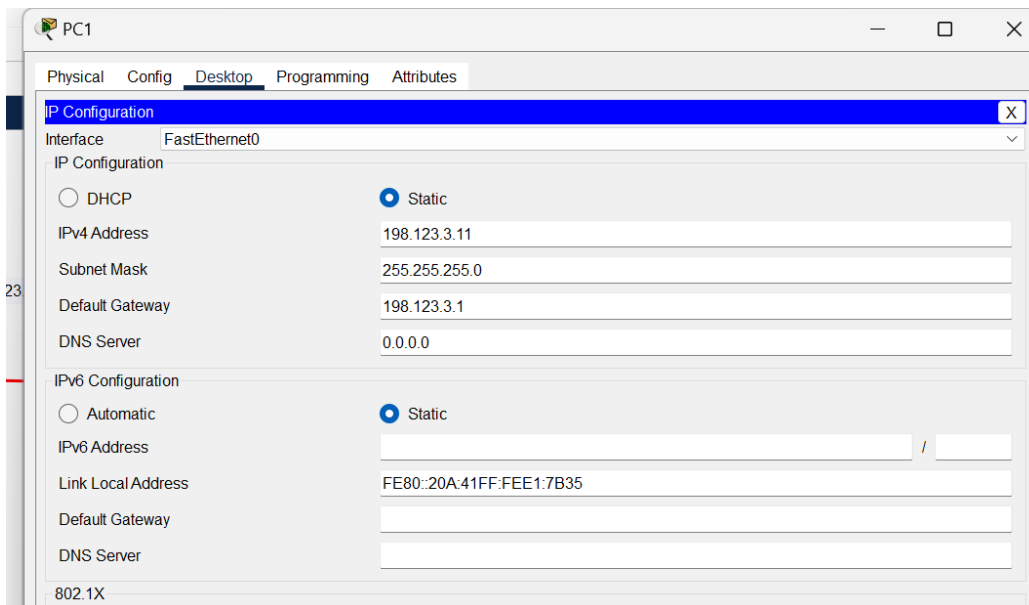
PC0 IPconfiguration



The screenshot shows the 'IP Configuration' window for PC0. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '198.123.1.11', 'Subnet Mask' is '255.255.255.0', 'Default Gateway' is '198.123.1.1', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' is empty, and the 'Link Local Address' is 'FE80::210:11FF:FEBA:769B'.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	198.123.1.11
Subnet Mask	255.255.255.0
Default Gateway	198.123.1.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::210:11FF:FEBA:769B

PC1 IPconfiguration

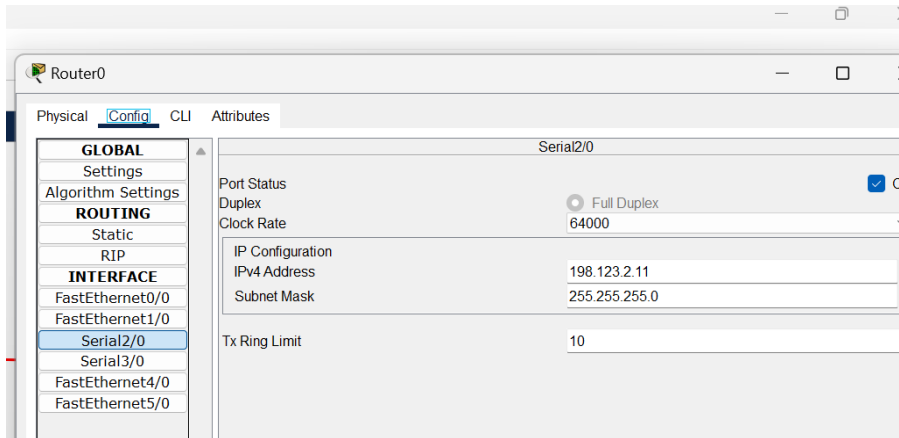


The screenshot shows the 'IP Configuration' window for PC1. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '198.123.3.11', 'Subnet Mask' is '255.255.255.0', 'Default Gateway' is '198.123.3.1', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' is empty, and the 'Link Local Address' is 'FE80::20A:41FF:FEE1:7B35'.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	198.123.3.11
Subnet Mask	255.255.255.0
Default Gateway	198.123.3.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::20A:41FF:FEE1:7B35

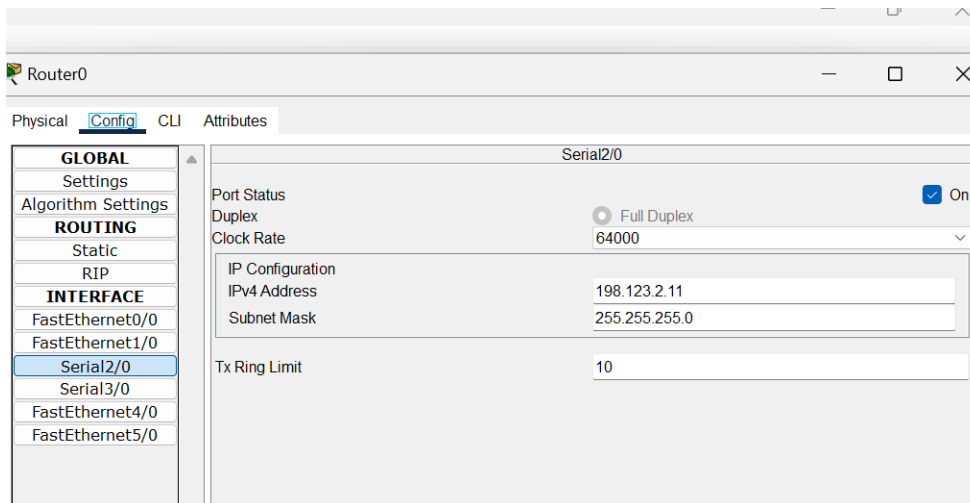
Router Configuration:-

At ROUTER 0:-



The screenshot shows the configuration window for Router0. The 'Config' tab is selected. On the left, the 'INTERFACE' section is expanded, and 'Serial2/0' is selected. The main area displays the configuration for Serial2/0:

Serial2/0	
Port Status	☒ On
Duplex	☐ Full Duplex
Clock Rate	64000
IP Configuration	
IPv4 Address	198.123.2.11
Subnet Mask	255.255.255.0
Tx Ring Limit	10



This screenshot is identical to the one above, showing the configuration window for Router0 with the 'Serial2/0' interface selected. The configuration details are the same:

Serial2/0	
Port Status	☒ On
Duplex	☐ Full Duplex
Clock Rate	64000
IP Configuration	
IPv4 Address	198.123.2.11
Subnet Mask	255.255.255.0
Tx Ring Limit	10

AT ROUTER 1:-

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.5806.D322

IP Configuration

IPv4 Address 198.123.3.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 198.123.2.12

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Commands at router 0:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, ch
exit
Router(config)#router bgp 100
Router(config-router)#network 198.123.1.0
Router(config-router)#network 198.123.2.0
Router(config-router)#neighbor 198.123.2.12 remote-as 200
Router(config-router)#neighbor 198.123.3.11 remote-as-200
^
```

% Invalid input detected at '^' marker.

```
Router(config-router)#neighbor 198.123.3.11 remote-as 200
Router(config-router)#exit
```

Commands at router 1:

```

Router(config-if)#exit
Router(config)#router bgp 200
Router(config-router)#network 198.123.2.0
Router(config-router)#network 198.123.3.0
Router(config-router)#neighbor 198.123.2.11 remote-as 100
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 198.123.2.11 Up
nennen
Router(config-router)#neighbhor 198.123.1.11 remote-as 100
^
% Invalid input detected at '^' marker.

Router(config-router)#neighbor 198.123.1.11 remote-as 100
Router(config-router)#exit

```

Final result

